Ma 3 PS1

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§1 Prove Boole's Law. Express both sides in terms of indicator functions and use the fact that expectation is a positive linear operator.

We can define an indicator variable as

$$I_{E_i}(\omega) = \begin{cases} 1 & \text{if } \omega \in E_i \\ 0 & \text{if } \omega \notin E_i \end{cases}$$

We know $P(A) = E[I_A]$. Using this, the two sides of Boole's inequality become

$$P\left(\bigcup_{i=1}^{n} E_i\right) = E\left[I_{\bigcup_{i=1}^{n} E_i}\right], \qquad \sum_{i=1}^{n} P(E_i) = \sum_{i=1}^{n} E[I_{E_i}]$$

We can now create an inequality between the indicator random variables. Idea: for every outcome ω in the sample space,

$$I_{\bigcup_{i=1}^{n} E_i}(\omega) \leq \sum_{i=1}^{n} I_{E_i}(\omega)$$

This can be broken down into two cases:

- Case 1: $\omega \notin \bigcup_{i=1}^{n} E_i$. Then $I_{\bigcup_{i=1}^{n} E_i}(\omega) = 0$ and $\sum_{i=1}^{n} I_{E_i}(\omega) \geq 0$, so the inequality works.
- Case 2: $\omega \in \bigcup_{i=1}^{n} E_i$. Then $I_{\bigcup_{i=1}^{n} E_i}(\omega) = 1$. Since ω belongs to at least one of the E_i, $\sum_{i=1}^{n} I_{E_i}(\omega) \geq 1$, so the inequality works again.

Next we can use the two properties of expectations:

1. Positivity: if $X(\omega) \leq Y(\omega)$ for all ω, then $E[X] \leq E[Y]$.
2. Linearity: $E\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} E[X_i]$.

Combining these with the inequality and the identity $P(A) = E[I_A]$ from above:

$$\boxed{P\left(\bigcup_{i=1}^{n} E_i\right) = E\left[I_{\bigcup_{i=1}^{n} E_i}\right] \leq E\left[\sum_{i=1}^{n} I_{E_i}\right] = \sum_{i=1}^{n} E[I_{E_i}] = \sum_{i=1}^{n} P(E_i)}$$

where the inequality comes from positivity and the second equality comes from linearity.

§2 An urn contains thirty-five red balls and sixty-five white balls. Eleven balls are drawn out and discarded, and a twelfth ball is drawn. What is the probability that the twelfth ball is red? Explain your reasoning.

Let p_k be the probability that the kth draw is red. We will show by induction that

$$p_k = \frac{35}{100} \quad \text{for every } k = 1, 2, \ldots, 12.$$

Base case: The first draw is red with probability

$$p_1 = \frac{35}{100}.$$

Induction: Assume for some $k \geq 1$ that

$$p_k = \frac{35}{100}.$$

For the $(k+1)$st draw, right before that draw, k balls have been removed, so there are $100 - k$ balls left. Let R_k be the number of red balls left at that moment. Then

$$\mathbb{P}((k+1)\text{st is red} \mid R_k) = \frac{R_k}{100 - k}.$$

Averaging over all possible values of R_k:

$$p_{k+1} = \mathbb{E}\left[\frac{R_k}{100 - k}\right] = \frac{\mathbb{E}[R_k]}{100 - k}.$$

So now we need to find $\mathbb{E}[R_k]$. Each of the first k draws removes a red ball with probability $p_1, p_2, \ldots, p_k$, so the expected number of red balls removed in the first k draws is

$$p_1 + p_2 + \cdots + p_k.$$

So

$$\mathbb{E}[R_k] = 35 - (p_1 + p_2 + \cdots + p_k).$$

Using the inductive hypothesis that $p_i = \frac{35}{100}$ for all $i \leq k$,

$$\mathbb{E}[R_k] = 35 - k \cdot \frac{35}{100} = 35\left(1 - \frac{k}{100}\right) = 35 \cdot \frac{100 - k}{100}.$$

Substitute this into the formula for p_{k+1}:

$$p_{k+1} = \frac{\mathbb{E}[R_k]}{100 - k} = \frac{35 \cdot \frac{100-k}{100}}{100 - k} = \frac{35}{100}.$$

So $p_{k+1} = \frac{35}{100}$, completing the induction.

$$\mathbb{P}(12\text{th ball is red}) = p_{12} = \frac{35}{100} = \boxed{\frac{7}{20}}$$

§3 Three coins look identical, except one is a regular heads/tails coin, one is heads/heads, and one is tails/tails.

We have coins HT, HH, and TT. Picking a coin at random, each has probability $\frac{1}{3}$, and then we pick a random side to view.

§3.1 If we randomly pick a coin and then choose a random side to look at, what is the probability of the event The observed side shows heads and the coin is the heads/heads coin?

We want

$$\mathbb{P}(\text{Heads observed and coin is } HH).$$

$$\mathbb{P}(HH) \cdot \mathbb{P}(\text{Heads observed} \mid HH) = \frac{1}{3} \cdot 1 = \frac{1}{3}.$$

$$\boxed{\frac{1}{3}}$$

§3.2 If we see that the observed side shows heads, what is the probability that it is the heads/heads coin?

We want

$$\mathbb{P}(HH \mid \text{Heads observed}).$$

Use Bayes' rule:

$$\mathbb{P}(HH \mid H) = \frac{\mathbb{P}(H \mid HH)\mathbb{P}(HH)}{\mathbb{P}(H)}.$$

To find $\mathbb{P}(H)$:

$$\mathbb{P}(H \mid HH) = 1, \quad \mathbb{P}(H \mid HT) = \frac{1}{2}, \quad \mathbb{P}(H \mid TT) = 0,$$

so

$$\mathbb{P}(H) = 1 \cdot \frac{1}{3} + \frac{1}{2} \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} = \frac{1}{3} + \frac{1}{6} = \frac{1}{2}.$$

So

$$\mathbb{P}(HH \mid H) = \frac{1 \cdot \frac{1}{3}}{\frac{1}{2}} = \frac{2}{3}.$$

$$\boxed{\frac{2}{3}}$$

§4 A crime is committed by one of two suspects, A and B. Initially, there is equal evidence against both of them. In further investigation at the crime scene, it is found that the guilty party had a blood type found in 10% of the population. Suspect A matches this blood type, whereas the blood type of Suspect B is unknown. Assume that the suspects' blood types are independent of each other and match the population distribution.

Let P_A and P_B be events that A and B are guilty respectively so $\mathbb{P}(P_A) = \mathbb{P}(P_B) = \frac{1}{2}$.
Let T be the event that a person matches the crime scene blood type so $\mathbb{P}(T) = 0.1$.

§4.1 Given this new information, what is the probability that A is the guilty party?

Let E be the new evidence where the guilty party matches the crime scene type and A matches so for example E occurs automatically if P_A is true and occurs under P_B only if T_B is true.

Therefore,

$$\mathbb{P}(E \mid P_A) = 1, \qquad \mathbb{P}(E \mid P_B) = \mathbb{P}(T_B) = 0.1.$$

By Bayes' rule,

$$\mathbb{P}(P_A \mid E) = \frac{\mathbb{P}(E \mid P_A)\mathbb{P}(P_A)}{\mathbb{P}(E \mid P_A)\mathbb{P}(P_A) + \mathbb{P}(E \mid P_B)\mathbb{P}(P_B)}.$$

$$\mathbb{P}(P_A \mid E) = \frac{1 \cdot \frac{1}{2}}{1 \cdot \frac{1}{2} + 0.1 \cdot \frac{1}{2}} = \frac{\frac{1}{2}}{\frac{1}{2} + \frac{1}{20}} = \frac{\frac{1}{2}}{\frac{11}{20}} = \boxed{\frac{10}{11}}$$

§4.2 Given this new information, what is the probability that B's blood type matches that found at the crime scene?

Given that the blood types follow the poulation distribution and are independent, B matches the crime scene type with probability

$$\mathbb{P}(T_B) = 0.1.$$

§5 Let $F(x) = 1 - e^{-\alpha x^\beta}$ for $x \geq 0$, $\alpha > 0$, $\beta > 0$, and $F(x) = 0$ for $x < 0$. Show that F is a cumulative distribution function (CDF), and find the corresponding density.

We aim to satisfy:

1. F is non-decreasing.
2. $\lim_{x\to-\infty} F(x) = 0$.
3. $\lim_{x\to\infty} F(x) = 1$.

§5.1 Show F is a CDF

§5.1.1 $\lim_{x\to-\infty} F(x) = 0$

$\forall x < 0$ $F(x) = 0$ so

$$\lim_{x\to-\infty} F(x) = 0 \quad \checkmark$$

§5.1.2 $\lim_{x\to\infty} F(x) = 1$

Since $\alpha > 0$ and $\beta > 0$, as $x \to \infty$ we have $\alpha x^\beta \to \infty$ so $e^{-\alpha x^\beta} \to 0$. So

$$\lim_{x\to\infty} F(x) = \lim_{x\to\infty} \left(1 - e^{-\alpha x^\beta}\right) = 1 - 0 = 1 \quad \checkmark$$

§5.1.3 F is non decreasing

For $x < 0$, $F(x) = 0$. At $x = 0$, $F(0) = 1 - e^0 = 0$, so F is constant (hence non decreasing) on $(-\infty, 0]$. For $x \geq 0$:

$$F'(x) = \frac{d}{dx}\left(1 - e^{-\alpha x^\beta}\right) = \alpha\beta x^{\beta-1} e^{-\alpha x^\beta}$$

Since $\alpha > 0$, $\beta > 0$, $x^{\beta-1} \geq 0$ for $x \geq 0$, and $e^{-\alpha x^\beta} > 0$, $F'(x) \geq 0$ for all $x \geq 0$. Therefore F is non-decreasing on $[0, \infty)$.

Combined with the fact that F is constant at 0 for $x < 0$ and continuous at $x = 0$, F is non decreasing on all of $\mathbb{R}$. ✓

Therefore F is a valid CDF.

§5.2 Find the Corresponding Density

The PDF is $f(x) = F'(x)$. From above:

$$\boxed{f(x) = \begin{cases} \alpha\beta x^{\beta-1} e^{-\alpha x^\beta} & x \geq 0 \\ 0 & x < 0 \end{cases}}$$

§6 Let $f(x) = (1 + \alpha x)/2$ for $-1 \le x \le 1$ and $f(x) = 0$ otherwise, where $-1 \le \alpha \le 1$. Show that f is a density, and find the corresponding cumulative distribution function (CDF). Find the median of the distribution in terms of α. Recall that the median m is the number such that $F(m) = 1/2$.

§6.1 Show that f is a density.

We aim to satisfy:

1. $f(x) \ge 0$ for all x.
2. $\int_{-\infty}^{\infty} f(x)\,dx = 1$.

§6.1.1 $f(x) \ge 0$ for all x

For $x \notin [-1, 1]$, $f(x) = 0 \ge 0$. For $x \in [-1, 1]$, the minimum of $1 + \alpha x$ over this interval occurs at one of the endpoints. At $x = 1$, $1 + \alpha$; at $x = -1$, $1 - \alpha$. Since $-1 \le \alpha \le 1$, both $1 + \alpha \ge 0$ and $1 - \alpha \ge 0$. Since $1 + \alpha x$ is linear in x, it is non-negative on the entire interval, so $f(x) = (1 + \alpha x)/2 \ge 0$. ✓

§6.1.2 $\int_{-\infty}^{\infty} f(x)\,dx = 1$

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-1}^{1} \frac{1 + \alpha x}{2}\,dx = \frac{1}{2}\left[\int_{-1}^{1} 1\,dx + \alpha \int_{-1}^{1} x\,dx\right] = \frac{1}{2}(2 + 0) = 1 \quad ✓$$

Therefore f is a density.

§6.2 Find the CDF

For $x < -1$, $F(x) = 0$. For $x > 1$, $F(x) = 1$. For $-1 \le x \le 1$,

$$F(x) = \int_{-1}^{x} \frac{1 + \alpha t}{2}\,dt = \frac{1}{2}\left[t + \frac{\alpha t^2}{2}\right]_{-1}^{x} = \frac{1}{2}\left[\left(x + \frac{\alpha x^2}{2}\right) - \left(-1 + \frac{\alpha}{2}\right)\right]$$

$$= \frac{1}{2}\left[x + 1 + \frac{\alpha x^2}{2} - \frac{\alpha}{2}\right] = \frac{1}{2}\left[x + 1 + \frac{\alpha(x^2 - 1)}{2}\right]$$

$$\boxed{F(x) = \begin{cases} 0 & x < -1 \\ \frac{1}{2}\left[x + 1 + \frac{\alpha(x^2 - 1)}{2}\right] & -1 \le x \le 1 \\ 1 & x > 1 \end{cases}}$$

§6.3 Find the Median

The median m should satisfy $F(m) = \frac{1}{2}$, where $m \in [-1, 1]$.

$$\frac{1}{2}\left[m + 1 + \frac{\alpha(m^2 - 1)}{2}\right] = \frac{1}{2}$$

$$\alpha m^2 + 2m - \alpha = 0$$

If $\alpha = 0$, $2m = 0$ so $m = 0$. If $\alpha \ne 0$, this is a quadratic in m. Using quadratic formula:

$$m = \frac{-2 \pm \sqrt{4 + 4\alpha^2}}{2\alpha} = \frac{-1 \pm \sqrt{1 + \alpha^2}}{\alpha}$$

We need $m \in [-1, 1]$. For the + root, when $\alpha > 0$: $m = (-1 + \sqrt{1+\alpha^2})/\alpha$. Since $\sqrt{1+\alpha^2} > 1$, the numerator is positive, and $\sqrt{1+\alpha^2} < 1 + \alpha$ for $0 < \alpha \leq 1$, so $m < 1$. So this root is in $[-1, 1]$. The $-$ root yields $m = (-1 - \sqrt{1+\alpha^2})/\alpha < -1$, which is outside the interval. Same thing for $\alpha < 0$. Therefore,

$$\boxed{m = \frac{-1 + \sqrt{1+\alpha^2}}{\alpha}}$$

§7 Let X be the result of rolling an ordinary six-sided die. Find $E(X)$, $E(X^2)$, and $E(1/X)$.

§7.1 Find $E(X)$

$$E(X) = \frac{1}{6}\sum_{k=1}^{6} k = \frac{1}{6}(1+2+3+4+5+6) = \frac{21}{6} = \boxed{\frac{7}{2}}$$

§7.2 Find $E(X^2)$

$$E(X^2) = \sum_{k=1}^{6} k^2 \cdot \mathbb{P}(X=k) = \frac{1}{6}\sum_{k=1}^{6} k^2 = \frac{1}{6}\left(\frac{6(6+1)(2(6)+1)}{6}\right) = \boxed{\frac{91}{6}}$$

§7.3 Find $E(1/X)$

$$E\left(\frac{1}{X}\right) = \sum_{k=1}^{6} \frac{1}{k} \cdot \mathbb{P}(X=k) = \frac{1}{6}\left(1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\frac{1}{5}+\frac{1}{6}\right) = \boxed{\frac{49}{120}}$$

§8 Recall that for a discrete random variable X the expected value is $E(X) = \sum k \cdot P(X = k)$ over all values k that X can take. If X takes values on the positive integers, show that $E(X) = \sum_{k=1}^{\infty} P(X \geq k)$. Apply this result to find the expected value of a geometric random variable.

§8.1 Show that $E(X) = \sum_{k=1}^{\infty} P(X \geq k)$

Since $X \in \mathbb{Z}^+$ we can express a positive integer k as a sum of ones: $k = \sum_{j=1}^{k} 1$. Substituting this:

$$E(X) = \sum_{k=1}^{\infty} k \cdot P(X = k) = \sum_{k=1}^{\infty} \left(\sum_{j=1}^{k} 1 \right) P(X = k)$$

The summation region $\{(j, k) : 1 \leq j \leq k < \infty\}$. We can change the order of summation by summing over j first and then over $k \geq j$,

$$E(X) = \sum_{j=1}^{\infty} \sum_{k=j}^{\infty} P(X = k) = \sum_{j=1}^{\infty} P(X \geq j)$$

Which is

$$\boxed{E(X) = \sum_{k=1}^{\infty} P(X \geq k)}$$

§8.2 Apply this result to find the expected value of a geometric random variable

Let $X \sim \text{Geometric}(p)$, so $P(X = k) = (1-p)^{k-1}p$.

$$P(X \geq k) = \sum_{j=k}^{\infty} (1-p)^{j-1}p = p \cdot (1-p)^{k-1} \sum_{i=0}^{\infty} (1-p)^i = p \cdot (1-p)^{k-1} \cdot \frac{1}{p} = (1-p)^{k-1}$$

With $q = 1 - p$:

$$E(X) = \sum_{k=1}^{\infty} P(X \geq k) = \sum_{k=1}^{\infty} q^{k-1} = \sum_{i=0}^{\infty} q^i = \frac{1}{1-q} = \frac{1}{p}$$

since $|q| < 1$. So

$$\boxed{E(X) = \frac{1}{p}}$$